

SQL Part 10: Final Project, Build a Pet Shop Database from Scratch — Free Practice Guide

Beyond School · Ready to learn in a fun way! Anyone can learn — easy! Nothing to install, just a web browser.

Step 1: Open a free online SQL playground

- sqliteonline.com — choose "SQLite", paste, click Run.
- db-fiddle.com — pick "SQLite" as the database, paste and Run.
- sqlime.org — a simple SQLite editor in your browser.
- sqlbolt.com — free interactive lessons for extra practice.

Step 2: Paste this setup script and click Run

It creates the same library tables used in the video. If anything goes wrong, just run it again in a fresh playground.

```
-- Part 10 builds the pet shop from scratch. Run every statement below, in order.
```

Step 3: Run every query from the video, in order

Paste one at a time, click Run, and compare with the expected result.

Build: owners

```
CREATE TABLE owners (    owner_id INTEGER PRIMARY KEY,    name TEXT NOT NULL,    city TEXT,    phone TEXT UNIQUE); INSERT INTO owners VALUES    (1, 'Maya', 'Austin', '555-0101'), (2, 'Leo', 'Denver', '555-0102'),    (3, 'Priya', 'Austin', '555-0103'), (4, 'Sam', 'Boston', '555-0104'),    (5, 'Zoe', 'Denver', '555-0105');
SELECT * FROM owners;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: owners
```

Build: pets

```
CREATE TABLE pets (    pet_id INTEGER PRIMARY KEY,    name TEXT NOT NULL, species TEXT NOT NULL,    birth_year INTEGER,    owner_id INTEGER REFERENCES owners(owner_id)); INSERT INTO pets VALUES (1, 'Biscuit', 'Dog', 2019, 1),    (2, 'Whiskers', 'Cat', 2021, 1), (3, 'Rex', 'Dog', 2016, 2),    (4, 'Nemo', 'Fish', 2024, 3), (5, 'Luna', 'Cat', 2020, 3),    (6, 'Coco', 'Parrot', 2018, 4), (7,
```

```
'Pepper', 'Dog', 2022, 3);
SELECT * FROM pets;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: pets
```

Build: products

```
CREATE TABLE products (   product_id INTEGER PRIMARY KEY,   name TEXT NOT NULL, category TEXT,   price REAL CHECK (price > 0)); INSERT INTO products VALUES (1, 'Dog Food 5kg', 'Food', 25.0),   (2, 'Cat Food 2kg', 'Food', 15.0), (3, 'Squeaky Bone', 'Toy', 6.0),   (4, 'Fish Flakes', 'Food', 4.5), (5, 'Bird Cage', 'Home', 80.0),   (6, 'Cat Tree', 'Home', 60.0), (7, 'Laser Pointer', 'Toy', 8.0);
SELECT * FROM products;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: products
```

Build: sales + sale_items

```
CREATE TABLE sales (sale_id INTEGER PRIMARY KEY,     owner_id INTEGER NOT NULL REFERENCES owners(owner_id),     sale_date TEXT NOT NULL); CREATE TABLE sale_items (     sale_id INTEGER REFERENCES sales(sale_id),    product_id INTEGER REFERENCES products(product_id),    qty INTEGER DEFAULT 1 CHECK (qty > 0),   PRIMARY KEY (sale_id, product_id));
SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Expected result:
name
```

Fill: a week of sales

```
BEGIN;  INSERT  INTO  sales  VALUES  (1,  1,  '2026-03-01'),  (2,  2,  '2026-03-02'),      (3,  3,  '2026-03-02'), (4, 1, '2026-03-05'),    (5, 4, '2026-03-07'), (6, 3, '2026-03-09'); INSERT INTO sale_items VALUES (1, 1, 2), (1, 3, 1), (2, 1, 1),   (3, 4, 3), (3, 2, 1), (4, 2, 2), (4, 3, 2), (5, 5, 1),   (6, 1, 1), (6, 3, 1); COMMIT;
SELECT * FROM sale_items;
```

```
Error message (expected):
error
You can only execute one statement at a time.
```

```
Error message (expected):
error
no such table: sale_items
```

Q1: owners and their pets

```
SELECT o.name AS owner, p.name AS pet, p.species FROM pets AS p JOIN owners AS o ON p.owner_id = o.owner_id ORDER BY o.name, p.name;
```

```
Error message (expected):
error
no such table: pets
```

Q2: pets per species

```
SELECT species, COUNT(*) AS pets FROM pets GROUP BY species ORDER BY pets DESC, species;
```

```
Error message (expected):
error
no such table: pets
```

Q3: owners with no pets

```
SELECT o.name FROM owners AS o LEFT JOIN pets AS p ON o.owner_id = p.owner_id WHERE p.pet_id IS NULL;
```

```
Error message (expected):
error
no such table: owners
```

Q4: total of each sale

```
SELECT s.sale_id, s.sale_date,           SUM(si.qty * pr.price) AS total FROM sales AS s JOIN sale_items AS si ON s.sale_id = si.sale_id JOIN products AS pr ON si.product_id = pr.product_id GROUP BY s.sale_id;
```

```
Error message (expected):
error
no such table: sales
```

Q5: best-selling products

```
SELECT pr.name, SUM(si.qty) AS sold FROM sale_items AS si JOIN products AS pr ON si.product_id = pr.product_id GROUP BY pr.name ORDER BY sold DESC, pr.name LIMIT 3;
```

```
Error message (expected):
error
no such table: sale_items
```

Q6: rank the customers

```
SELECT o.name, SUM(si.qty * pr.price) AS spent,    RANK() OVER (ORDER BY SUM(si.qty * pr.price) DESC) AS rnk FROM owners AS o JOIN sales AS s ON o.owner_id = s.owner_id JOIN sale_items AS si ON s.sale_id = si.sale_id JOIN products AS pr ON si.product_id = pr.product_id GROUP BY o.name;
```

```
Error message (expected):
error
no such table: owners
```

Q7: running revenue

```
WITH daily AS (    SELECT s.sale_date, SUM(si.qty * pr.price) AS revenue    FROM sales AS s    JOIN sale_items AS si ON s.sale_id = si.sale_id    JOIN products AS pr ON si.product_id = pr.product_id    GROUP BY s.sale_date) SELECT sale_date, revenue,    SUM(revenue) OVER (ORDER BY sale_date) AS so_far FROM daily;
```

```
Error message (expected):
error
no such table: sales
```

Q8: products never sold

```
SELECT name, price FROM products AS p WHERE NOT EXISTS (    SELECT 1 FROM sale_items AS si    WHERE si.product_id = p.product_id);
```

```
Error message (expected):
error
no such table: products
```

Q9: price labels

```
SELECT name, price,    CASE WHEN price >= 50 THEN 'Big buy'         WHEN price >= 10 THEN 'Regular'       ELSE 'Small treat' END AS label FROM products ORDER BY price DESC;
```

```
Error message (expected):
error
no such table: products
```

Q10: above-average spenders (view)

```
CREATE VIEW vip AS WITH spend AS (    SELECT s.owner_id, SUM(si.qty * pr.price) AS spent    FROM sales AS s    JOIN sale_items AS si ON s.sale_id = si.sale_id    JOIN products AS pr ON
```

```
si.product_id = pr.product_id   GROUP BY s.owner_id) SELECT o.name, sp.spent FROM spend AS sp JOIN owners AS o ON sp.owner_id = o.owner_id WHERE sp.spent > (SELECT AVG(spent) FROM spend);
SELECT * FROM vip ORDER BY spent DESC;
```

```
Error message (expected):
error
no such table: main.sales
```

Bonus: an index

```
CREATE INDEX idx_sales_owner ON sales(owner_id); CREATE INDEX idx_pets_owner ON pets(owner_id);
EXPLAIN QUERY PLAN SELECT * FROM pets WHERE owner_id = 3;
```

```
Error message (expected):
error
no such table: main.sales
```

```
Error message (expected):
error
no such table: pets
```

Step 4: Challenges (try first!)

C1. Add a grooming appointments table for pets, then find which pet has the most appointments.
C2. Total revenue per product category.
C3. Which owners in Austin bought something?

Answers

C1. Add a grooming appointments table for pets, then find which pet has the most appointments.

```
CREATE TABLE appointments (appt_id INTEGER PRIMARY KEY, pet_id INTEGER REFERENCES pets(pet_id), appt_date TEXT, service TEXT);
INSERT INTO appointments VALUES (1, 1, '2026-03-03', 'Bath'), (2, 1, '2026-03-10', 'Nail trim'), (3, 5, '2026-03-04', 'Bath');
SELECT p.name, COUNT(*) AS visits FROM appointments AS a JOIN pets AS p ON a.pet_id = p.pet_id GROUP BY p.name ORDER BY visits DESC LIMIT 1;
```

```
Error message (expected):
error
no such table: pets
```

C2. Total revenue per product category.

```
SELECT pr.category, SUM(si.qty * pr.price) AS revenue FROM sale_items AS si JOIN products AS pr ON si.product_id = pr.product_id GROUP BY pr.category ORDER BY revenue DESC;
```

```
Error message (expected):
error
no such table: sale_items
```

C3. Which owners in Austin bought something?

```
SELECT DISTINCT o.name FROM owners AS o JOIN sales AS s ON o.owner_id = s.owner_id WHERE o.city = 'Austin';
```

```
Error message (expected):
error
no such table: owners
```

Keep going!

Watch the whole free SQL course on Beyond School, starting with SQL Part 1: https://youtu.be/72JvYu2OtiA
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